Mapping phone reception on the CFB Wainwright ranges

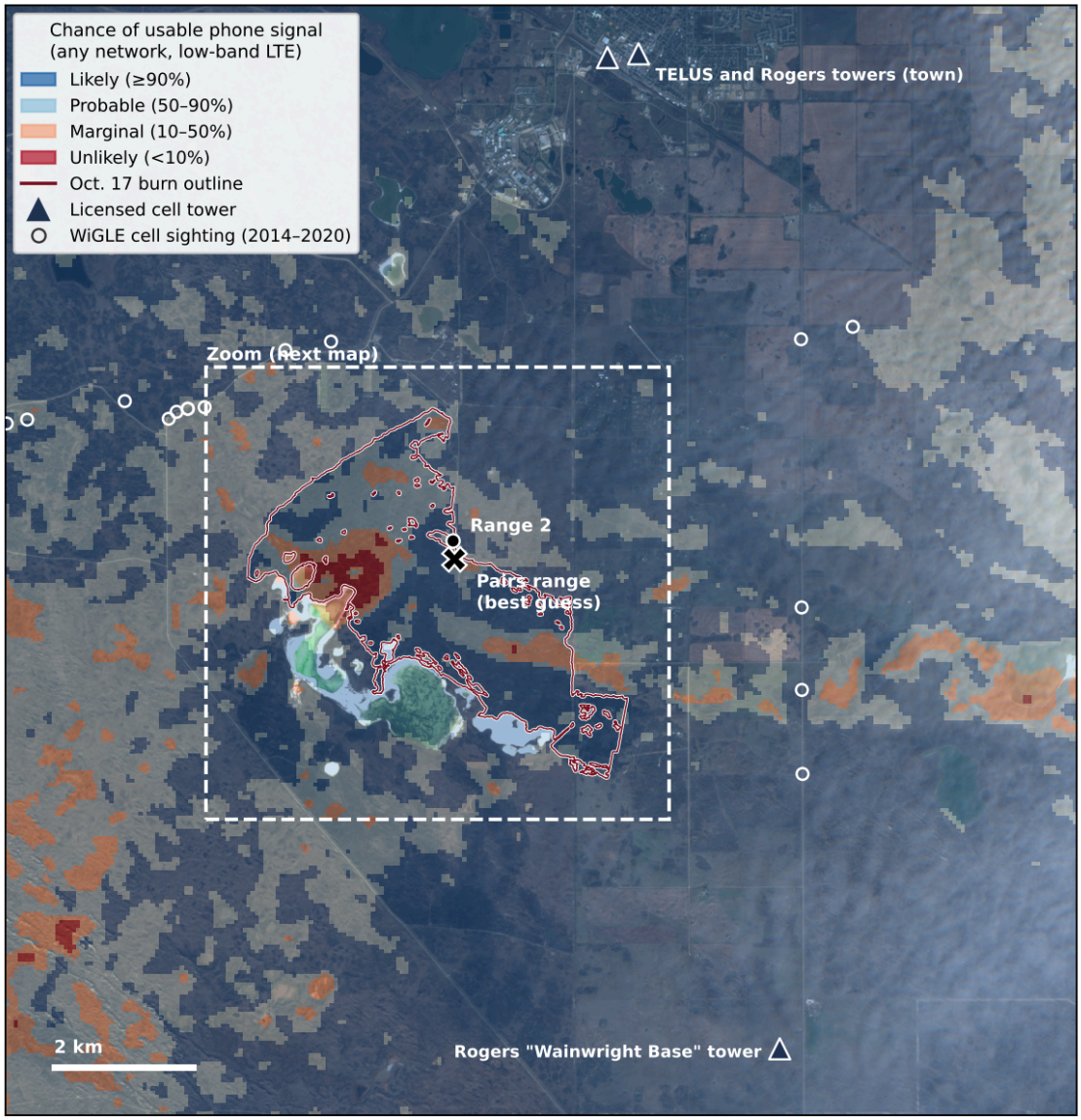
Kevin J.S. Duska Jr. · Oct. 29, 2026

Could the soldiers on the pairs range have made a call or got a text on the afternoon of Oct. 17? Nobody has published measured coverage for the training area, so I built a model. It uses the licensed cell towers, their heights, and the terrain between them and the ranges. It estimates the chance of a usable signal across the area around the fire, with confidence bands, laid over the burn scar, the ranges and satellite imagery.

> **This is a model, not a measurement.** It shows where reception was likely or unlikely given the towers and the terrain. It doesn't show what any phone actually did on Oct. 17.

Bottom line

- **On Rogers, signal near the pairs range was probably usable.** Within 300 m of our best estimate of the pairs range, the median chance of a usable Rogers signal is about **85%**. The single point varies a lot over short distances, from 39% to 96%. Almost all of it comes from one tower, Rogers' "WAINWRIGHT BASE", 8.5 km to the south-southeast.
- **On TELUS, it was more doubtful.** TELUS's towers are in town and farther out, so the same area has a median chance of only about **30%**. Bell customers in this region typically ride the same towers as TELUS through the two companies' network-sharing arrangement, so the same likely applies to them.
- **A 911 call would have done better than a regular call or text.** Emergency calls connect through any network in range, so someone on TELUS or Bell could still reach 911 over Rogers. Ordinary calls, texts and data generally can't.
- **Coverage drops in low ground.** The low basin and lakes on the west and south sides of the burn are the weakest spots, especially on TELUS. About half the burned area is "likely" on Rogers, but only 4% is on TELUS.
- **The few real-world observations agree with the model.** All 15 places where WiGLE contributors' phones logged TELUS or Bell between 2014 and 2020 fall in areas the model rates at a 47% chance of a usable TELUS signal or better (median 86%).

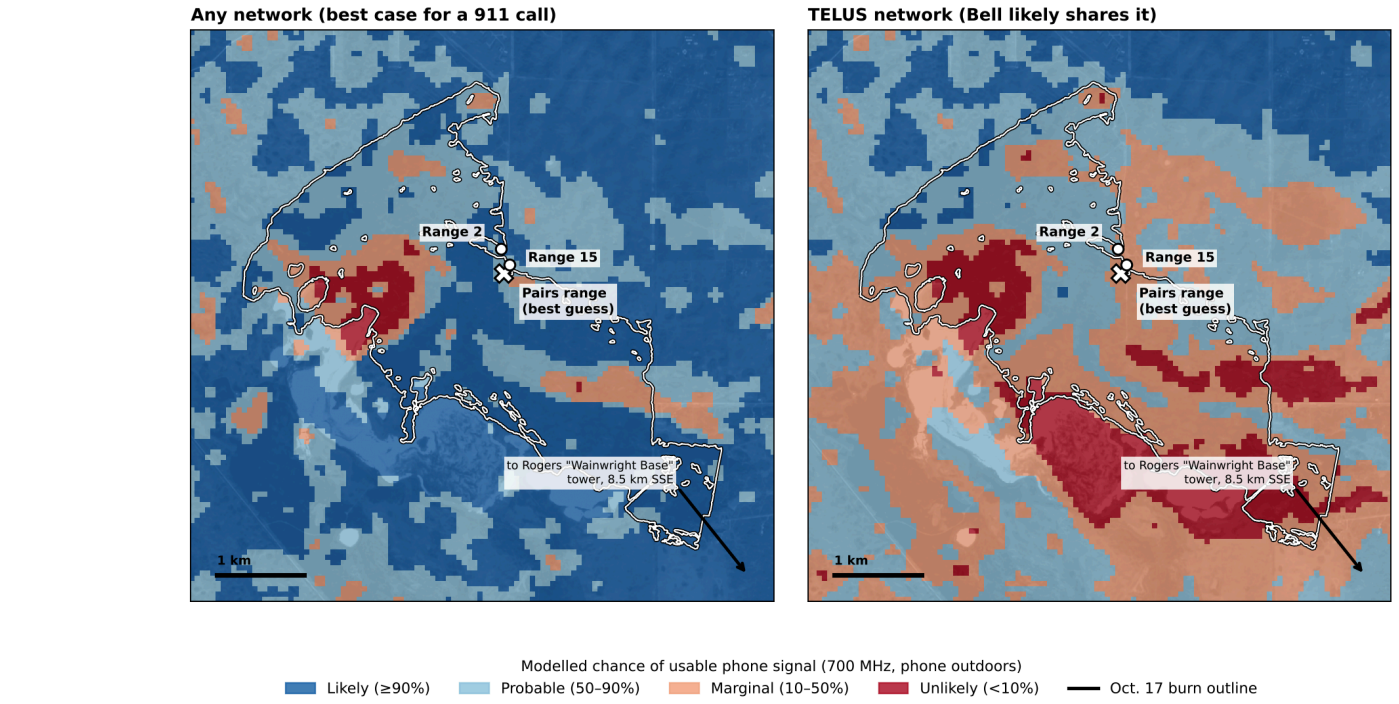


Modelled, not measured. Copernicus 30 m terrain; ISED tower sites and heights; 700 MHz; handset outdoors at 1.5 m; path-loss threshold 140 dB, 8 dB spread.
Base image: Sentinel-2, Oct. 19, 2026 (Copernicus). Burn outline and model: Prime Rogue Inc.

Modelled chance of a usable phone signal on any network, over Sentinel-2 imagery from Oct. 19, 2026. The burn outline is from 70 cm imagery taken Oct. 20. White circles are WiGLE cell sightings.

Close-up: the burn area

The same model, zoomed to the burn with stronger colours. On the left is any network, which is what matters for a 911 call. On the right is TELUS, which Bell customers likely share. The pairs range and Range 15 sit in a "likely" pocket on the left map but a "probable/marginal" patch on the right. The low basin west of the pairs range and the lakes along the burn's south side are the weakest areas on both.

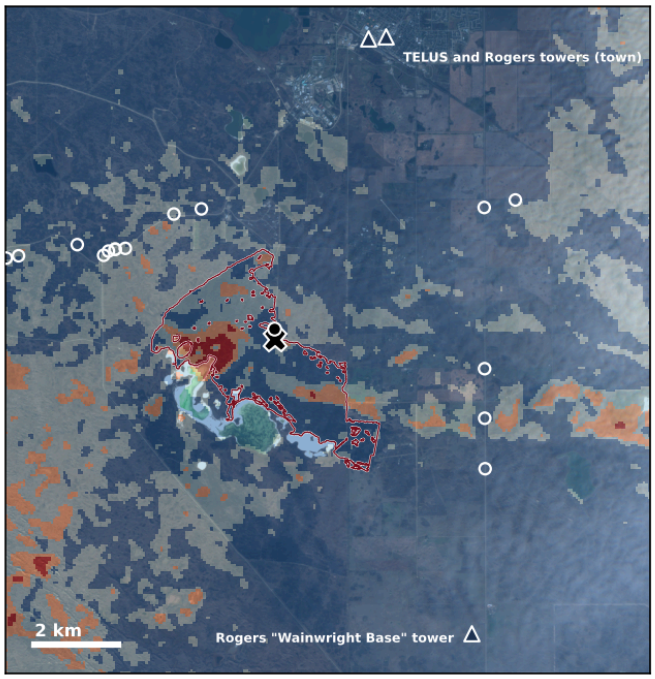


Model, not measurement. Threshold 140 dB, 8 dB spread. Base: Sentinel-2, Oct. 19, 2026 (greyscale). Burn outline and model: Prime Rogue Inc.

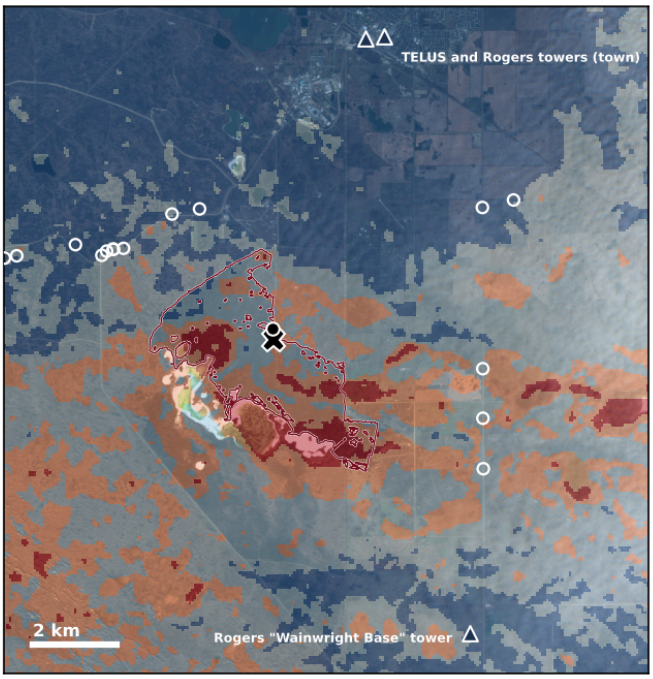
By network, and how sure I am

The top two maps split the result by network. The bottom two show how the "any network" picture changes under a cautious and an optimistic assumption about how much signal loss a phone can tolerate. Areas that stay blue in the cautious map are the most reliable "yes"; areas that stay red in the optimistic map are the most reliable "no".

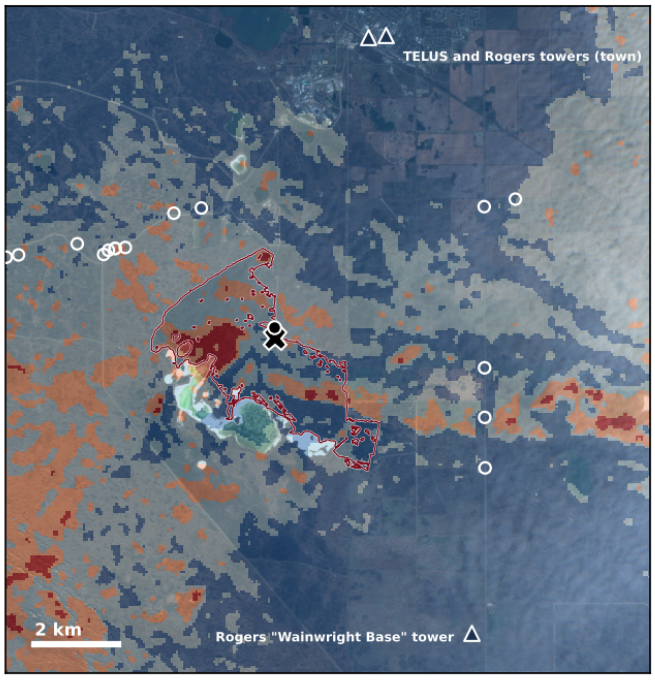
Rogers network (threshold 140 dB)



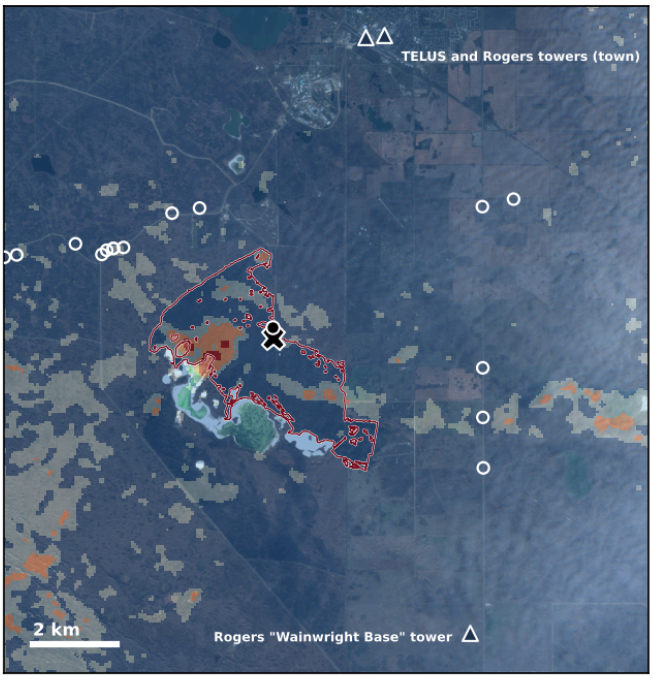
TELUS network, Bell likely shares it (140 dB)



Any network, cautious (135 dB)



Any network, optimistic (145 dB)



Likely (≥90%) Probable (50–90%) Marginal (10–50%) Unlikely (<10%)

Location (300 m around each point)	Rogers: median chance	TELUS/Bell: median chance	Any network, cautious to optimistic
Pairs range (our best estimate)	85%	32%	67% to 95%
Range 15 (official grid)	85%	31%	66% to 95%
Road head	85%	32%	66% to 95%

Range 2 (official grid)	92%	48%	79% to 98%

What it means for the story

- Phones were a plausible backup channel at the pairs range, mainly for anyone on Rogers or calling 911. They weren't a dependable one for someone on TELUS or Bell, and a short move in either direction could change the picture.
- On a military range, the normal channel for reporting an emergency is the unit's radio to Range Control, not a personal phone. Phones matter as a backup if radio failed or was overloaded, and for anyone who called 911 or family directly. Whether the first emergency call came through 911 is one of the questions put to Emergency Health Services Alberta.
- Even with signal, a warning had to be sent to a phone to arrive by phone. As our alerts analysis showed, the 2:14 p.m. wind warning wasn't pushed to phones as an emergency alert.
- What actually happened is a question for DND and the people who were there. Did anyone use a phone? Did calls connect? Were personal phones allowed or carried on the range during training?

Method and limits

Towers. Every licensed Rogers and TELUS macro cell site within about 25 km, from ISED's spectrum licence data dated Oct. 5, 2026, using each site's licensed structure height (antennas assumed at 90% of it). Small rooftop cells in town and fixed-wireless internet sites were left out. **Terrain.** The Copernicus 30 m digital surface model, which includes trees and buildings, with Earth curvature. **Signal.** 700 MHz (the long-range low band both carriers hold here). Loss is free-space or plane-earth, whichever is greater, plus diffraction over the terrain between tower and phone (knife-edge, main edge plus secondary edges) and an 8 dB allowance for clutter and the user's body. The phone is assumed outdoors at 1.5 m. **Usable signal** means total loss under 140 dB, with an 8 dB spread for local variation; 135 and 145 dB are the cautious and optimistic cases. Results are on a 60 m grid and smoothed over 180 m for the maps; values at named points are summarized over 300 m to reflect location uncertainty.

Not modelled: antenna direction and tilt, actual transmit power, network load during the emergency, phones inside vehicles or buildings, and any equipment changes since Oct. 17. WiGLE sightings can only confirm coverage where someone logged it; they can't confirm its absence. The pairs-range location is approximate.

Sources

- ISED, Spectrum Management System data: spectrum licence site data, Oct. 5, 2026
- Copernicus DEM GLO-30 (ESA / Copernicus, via AWS Open Data)
- Sentinel-2 L2A, Oct. 19, 2026 (Copernicus, via Earth Search)
- WiGLE.net crowdsourced cell sightings, searched Oct. 29, 2026

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